Name: Hemanth P

Regno: 192524297

# DAA LAB

## LAB EXPERIMENTS

## EXPERIMENT – 3

## BUBBLE SORT WITH BIG-O ANALYSIS

## AIM:

To implement Bubble Sort using Python and analyze its time complexity using Big-O notation.

## ALGORITHM:

1. Start the program.
2. Initialize an array of elements.
3. Compare adjacent elements in each pass.
4. Swap elements if they are in the wrong order.
5. Repeat until all elements are sorted.
6. Count the number of comparisons.
7. Display the sorted array and complexity.
8. Stop the program.

## PROCEDURE:

1. Declare an array with unsorted elements.
2. Use nested loops to compare adjacent elements.
3. Swap elements whenever the left element is greater than the right element.
4. Increment the comparison counter for every comparison.
5. Print the sorted array and total comparisons.
6. Analyze the time complexity.

## PYTHON CODE:

```python
a = [64, 34, 25, 12, 22]
n = len(a)
comparisons = 0

for i in range(n - 1):
    for j in range(n - 1 - i):
        comparisons += 1
        if a[j] > a[j + 1]:
            a[j], a[j + 1] = a[j + 1], a[j]

print("Sorted array:", a)
print("Comparisons:", comparisons)
print("Time Complexity: O(n^2)")
```

## SAMPLE OUTPUT:

```
Output

Sorted array: [12, 22, 25, 34, 64]
Comparisons: 10
Time Complexity: O(n^2)
```

## RESULT:

The Bubble Sort algorithm was implemented successfully. The array was sorted in ascending order, and its time complexity was analyzed as (O(n^2)).